Distant Family Outdoor Game Design - Bun Appétit

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Introduction

Family connection plays a vital role in emotional well-being, offering individuals a sense of identity, support, and continuity throughout life. Even as family structures evolve and geographic distance becomes more common, strong relationships between parents and adult children remain essential for mental health and intergenerational connection [11]. However, barriers such as time zone differences, schedule conflicts, chat topic preferences, or inability to find suitable topics to share lead to superficial interactions. While technologies like video chat and messaging help families stay in touch, they rarely replicate the emotional richness of face-to-face interaction [2]. Distant family relationships can suffer from a lack of meaningful connection, weakening emotional closeness over time.

To address this gap, we focus on the potential of joint activities. Joint activities, shared tasks that create a sense of collaboration and interdependence, can help distant family members feel more connected by giving their interactions purpose and structure. However, initiating these experiences across distance can be challenging due to the absence of shared routines, scheduling conflicts, and the discomfort of starting open-ended conversations [7]. By framing interaction around lightweight but goal-oriented tasks such as ingredient collection, photo sharing, or cooking memory posts, we lower the cognitive and emotional barriers that often prevent families from engaging together at a distance. Tools that support asynchronous participation, like shared diaries or turn-based play, allow users to contribute at their own pace, making it easier to stay connected despite schedule mismatches. These task-oriented designs reduce social friction, spark conversation, and support memory-making, all of which contribute to stronger emotional bonds over time [3].

Bun Appétit is a mobile app designed to transform everyday cooking into a collaborative, gamified experience for distant family members. By centering on joint recipe completion and playful documentation of food preparation, the app encourages participation in an emotionally rich, shared routine. A virtual kinkeeper, Nibbles the bunny, supports engagement through a friendly presence that adds warmth and continuity to the experience. Research in HCI and social computing supports this approach. Gamification increases motivation, symbolic agents create a sense of social presence, and embedding bonding into routine tasks fosters sustainable connection [6]. We hypothesize that *Bun Appétit* will deepen communication quality, stimulate shared memories, and enhance emotional closeness by making connection feel natural, fun, and purpose-driven.

To guide our evaluation of Bun Appétit, we focus on two primary research questions:

- **Can the gamification of joint activities facilitate emotional connection between distant family members?** This question explores whether features such as collaborative tasks, playful prompts, and progress tracking can not only encourage participation but also foster a sense of closeness and shared purpose.

- **How do users perceive the app's role in initiating and structuring interactions?** We investigate whether the app helps reduce the social friction often associated with starting conversations or planning shared activities, and whether it offers a meaningful and sustainable framework for remote connection.

To answer these questions, we conducted a four-day user study with 13 participants from 9 families, each of whom completed a full cycle of the game and provided qualitative reflections through a post-study questionnaire. Our preliminary findings suggest that (1) task-oriented framing lowered barriers to start and sustain joint activities. We also found that (2) the presence of a symbolic companion, Nibbles, contributed to emotional continuity and motivation, serving as a lightweight but meaningful social presence. Additionally, (3) activity-based design can effectively trigger and sustain interactions beyond the playtime. Finally, (4) participants highlighted the asynchronous game-play process to coordinate joint activities. These findings contribute to the broader game research and design community by demonstrating how narrative-driven, emotionally anchored, and task-based interaction models can transform routine activities like cooking into playful, collaborative experiences. Our work offers the potential of gamifying shareable routine experiences for game researchers and practitioners interested in supporting emotional connection and sustained engagement in geographically dispersed relationships.

Related work

2.1 Distant Family Communication and Joint Activity Challenges

Prior research has consistently demonstrated that maintaining emotional closeness across geographic distance presents enduring challenges for families, with barriers including time zone differences, limited shared context, and difficulties initiating meaningful conversations. Although technologies such as video calls and messaging applications provide mechanisms to sustain contact, they frequently fall short in supporting emotionally rich interactions or facilitating joint activities. Studies on parent-child relationships have further highlighted that discrepancies in expectations and interactional styles can compound the difficulties of remote communication. Research on intergenerational ties underscores the importance of maintaining strong parent-child bonds for individual well-being and family solidarity. However, the successful coordination of shared activities is rarely spontaneous and often depends on a family member assuming responsibility for organizing, motivating, and sustaining engagement, a relational function conceptualized in the family studies literature as *kinkeeping*.

2.2 Digital Kinkeeping

Kinkeeping refers to the relational labor frequently undertaken by specific family members, most often women, to maintain familial bonds and ensure regular interaction. While this role is vital for sustaining family cohesion, it can impose a disproportionate emotional and logistical burden on the individual kinkeepers. Recent scholarship has examined how digital technologies, including family group chats and social media platforms, can help distribute kinkeeping responsibilities more equitably across family members, thus reducing individual stress and creating a more collective model of relational maintenance. Additionally, research in human-computer interaction has shown that symbolic companions and virtual agents can enhance emotional presence, provide social cues, and foster a sense of connection in mediated settings. However, few studies have investigated how symbolic agents might function as shared emotional anchors within family systems, supporting or augmenting kinkeeping practices. Addressing this question is critical for informing the design of systems that aim to motivate and sustain family engagement over time.

2.3 Gamification and Emotional Engagement in Family Technologies

Gamification has emerged as a widely studied strategy for increasing user engagement, motivation, and enjoyment across a range of domains. Within the context of family technologies, gamified systems have demonstrated promise in sustaining participation and fostering a sense of shared achievement. Prior work on cooperative gameplay has further illustrated that asymmetric roles can promote intergenerational co-play and improve the quality of family communication by balancing abilities and engagement across participants. Despite these advances, much of the existing research has focused on entertainment-centered applications, with comparatively little attention given to the potential of gamification to enrich everyday family routines. The present study builds on this body of work by examining how gamified, task-oriented designs can reduce communication barriers, lower the burden of relational coordination, and foster emotional closeness between geographically dispersed family members.

Game Design

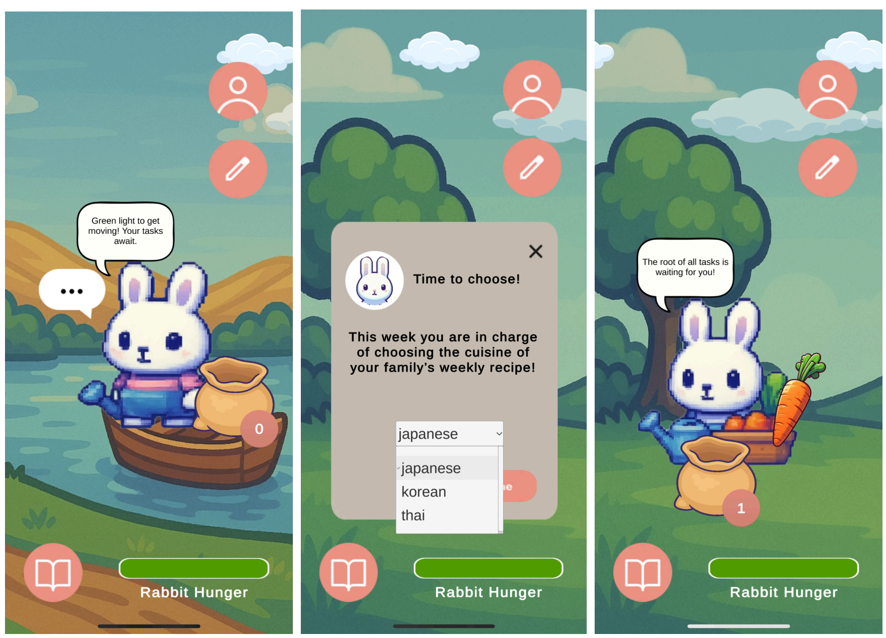
In this section, we present the overall game design of Bun Appétit. The app guides users through a structured cycle of collaboration, centered on cooking and memory-sharing. The interaction flow proceeds as follows:

> (1) Users begin in the Garden, where Nibbles, a virtual companion, prompts them to vote on a recipe attribute they've been assigned: cuisine, main ingredient, or dish type.
>
> (2) Once all users in the family have submitted their votes, a recipe is generated based on the selected combination of attributes.
>
> (3) Each user receives a set of individual ingredient tasks derived from the recipe. They complete these by photographing the ingredients, which are verified using OpenAI's GPT-4 Vision model.
>
> (4) When a task is completed and validated, the user earns carrot rewards, which they can then use to feed Nibbles.
>
> (5) After all family members complete their respective tasks, the recipe is marked as complete and moved into the RecipeBook, where it can be viewed at any time.

(6) Users can then contribute to the Memory Collage by uploading a photo of their final dish and writing a short diary entry. Other family members can browse the carousel of submissions, react with emojis, and leave comments to reflect on the shared experience.

This gameplay loop transforms a routine activity (preparing a meal) into a collaborative digital ritual that fosters emotional connection. In the following subsections, we introduce the core modules that make up the Bun Appétit gameplay experience and explain how each contributes to fostering connection between distant family members. We describe the Garden, which features Nibbles the virtual pet as a persistent emotional companion; the Tasks module, which guides users through collaborative steps; the Recipe Book, which visualizes progress and shared goals; and the Memory Collage, where families reflect on their joint accomplishments. Together, these modules integrate playful interaction, symbolic presence, and asynchronous participation to support emotionally meaningful engagement, with the backend architecture (built on Django) providing the foundation for real-time data synchronization, media handling, and task verification across the system.

Garden

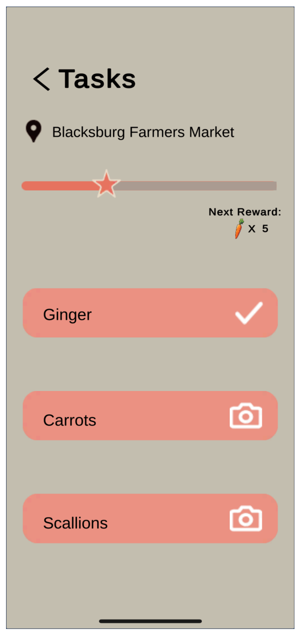


The Garden serves as the main home screen of Bun Appétit and acts as the emotional hub of the experience, with backend support handling user profiles, Nibbles' state, and recipe voting status to ensure synchronized progress across family members. It features Nibbles, a persistent virtual pet who functions as the family's symbolic kinkeeper. Nibbles delivers context-aware prompts through a speech bubble, which updates based on task progress, memory submissions, rabbit hunger, or whether a new recipe needs to be selected.

When no recipe is currently in progress, a speech bubble with three dots appears. Tapping on it opens a prompt for the user to vote on one of the three recipe-defining attributes they are assigned: cuisine, main ingredient, or dish type. Once all family members have submitted their votes, the system assigns a collaborative recipe based on the selected combination.

The Garden also facilitates user navigation. Tapping the avatar icon brings users to their profile, the pencil icon opens the task view, and the book icon leads to the RecipeBook. The background changes each time the user opens the app, creating a sense of freshness and liveliness. Users can also feed Nibbles by dragging carrots from the visible bag, which reflects how many carrots they currently have.

Tasks



The Tasks module breaks down a family's assigned recipe into interactive steps, turning ingredient collection into a playful, goal-oriented experience. At the top of the screen, the app displays a recommended local farmers market based on the user's location, encouraging real-world engagement with the recipe process. Below that, a progress bar shows how many tasks have been completed and how many carrots will be earned as a reward for the next completed task.

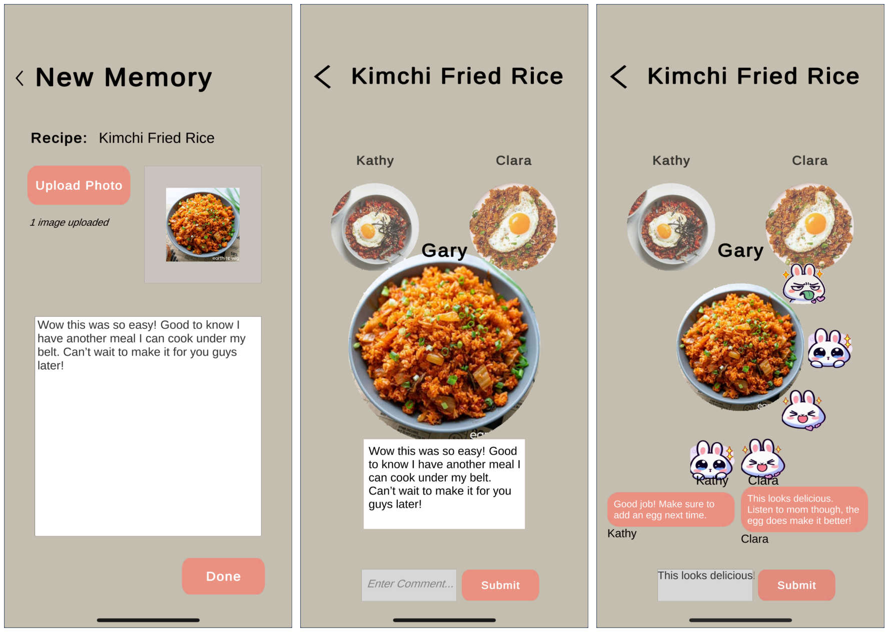
Each task corresponds to an ingredient from the current recipe and is shown with a camera icon. Users can tap the camera icon to photograph the ingredient, and the image is verified using OpenAI's GPT-4 Vision model to determine whether it matches the requested ingredient, with the backend managing image uploads, calling the GPT-4 model, and storing verification results for each user. When a submission is successfully validated, the task is marked complete and the user receives a carrot reward. When all users in the family complete their assigned tasks, the recipe is officially marked as “Complete.” This module helps anchor remote collaboration in tangible, shareable actions while keeping users motivated through visual feedback and incremental rewards.

Recipe Book

The Recipe Book module helps families keep track of their collaborative cooking progress and explore potential future meals. It is divided into two tabs: Family Recipes and Locked Recipes. The Family Recipes tab displays meals that are either currently in progress or fully completed, with backend systems tracking recipe states, task completion data, and storing family cooking history. If a user taps on a recipe marked “In Progress,” they are taken directly to the recipe page, which includes the ingredient list, cooking instructions, and a shortcut to the corresponding task list. Completed recipes link to their respective Memory Collage module.

The Locked Recipes tab contains meals that have not yet been unlocked. These recipes are potential candidates that could become available in the future through the family’s voting system. They are displayed visually but are not clickable, serving as a teaser for what families might cook together next. This layout encourages progress, discovery, and a sense of shared ownership over the recipe journey.

Memory Collage

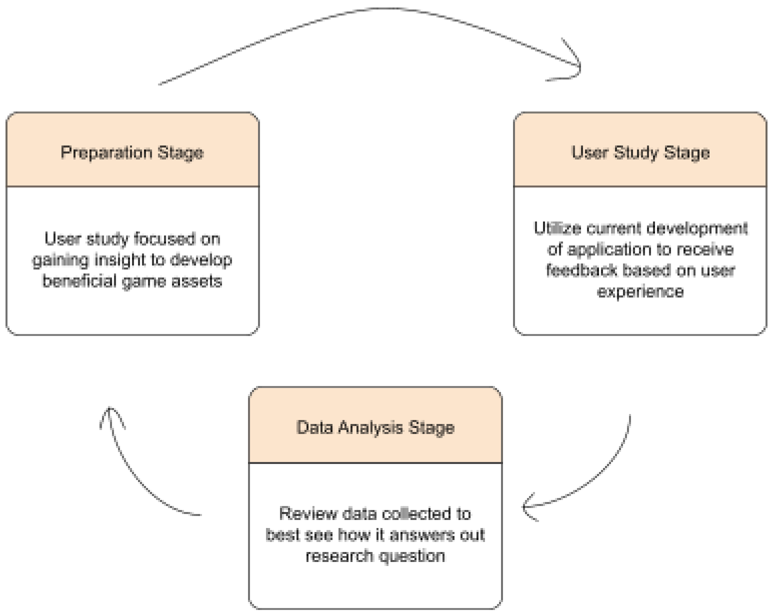


The Memory Collage module is where family members reflect on their shared cooking experiences by uploading and browsing finished dish memories. At the top of the screen, users can either add a new memory entry or navigate back to the corresponding recipe. The center of the screen features a rotating carousel of memory orbs, each representing a photo and diary entry submitted by a family member.

Each orb displays the uploaded image of a completed dish. The carousel layout is designed so that all family members' dishes are visible on screen at the same time, allowing users to appreciate everyone's contributions at a glance. Tapping on an orb expands it to show the full diary entry detailing that user's cooking experience. When not selected, users can still interact with the orb by choosing one of three emoji reactions displayed on its edge or by submitting a written comment. This lightweight interaction invites playful feedback and emotional response from other family members. The carousel can be navigated by swiping left or right to focus on different entries. Adding a new memory involves uploading a photo and writing a short personal note, turning the cooking task into a shared storytelling opportunity that strengthens long-distance bonds, while the backend handles media storage, diary entry records, and synchronization of emoji reactions and comments across devices.

Methodology

To evaluate how Bun Appétit supports emotional connection between distant family members, we conducted a four-day diary study with families using our prototype app. This section outlines our study design, including participant recruitment, data collection procedures, and qualitative analysis methods. Through a combination of app usage and a reflection done by participants, we observed how our participants experienced connection throughout the gameplay cycle.



Preparation Stage

We recruited participants directly through the authors' networks and invited them to participate in the study together with their distant family members. Eligible participants were asked to involve at least one family member living in a separate household, preferably at a geographically distant location, to match the intended use context of the application.

Each participating family pair or group was instructed to engage with the application over a four-day period, during which they completed one full cycle of the app's collaborative features. This included voting on a recipe-defining attribute (cuisine, main ingredient, or dish type), receiving and completing ingredient tasks, unlocking the recipe, contributing a memory entry with a dish photo and reflective note, and responding to at least one other family member's memory through a reaction or comment.

This protocol was designed to ensure participants experienced the complete interaction flow of *Bun Appétit*, enabling them to provide feedback on both individual engagement and the quality of family connection facilitated by the system.

User Study Stage

The user study was conducted by first providing interested participants with background information about the application and instructions on how to use it. Participants received a document detailing how to navigate the app, including how to create an account, set up their family group using a unique code, vote on a recipe attribute, complete assigned tasks, create a memory entry, and interact with family members through shared content and feedback.

Importantly, while participants were given a clear explanation of the app's features and interaction flow, they were **not informed of the specific goals of the study** in order to avoid introducing bias and to allow for objective observation of their natural user experience. Furthermore, participants were not provided with strict guidelines on when or how to use the app; they were encouraged to freely explore the system and engage with it in ways that felt natural and meaningful to them. This approach allowed participants to experience the app in a flexible and self-directed manner, reflecting realistic patterns of use.

After completing the four-day study, participants were asked to provide feedback through a post-study questionnaire. The feedback form was divided into two sections: family background and user experience.

In the **family background section**, we aimed to understand who participants used the app with and how it supported family communication. Questions included: how often they typically communicated with these family members (e.g., more often, less often, via calls or texts), how they previously engaged in food journaling or sharing cooking experiences, and what challenges they encountered when attempting remote joint activities (e.g., cooking together, sharing school or travel updates, sharing meals).

In the **user experience section**, we sought to capture participants' overall impressions of the app and identify any challenges they faced. Questions addressed participants' general experience, which features they found engaging or motivating, challenges encountered during family interaction, suggestions for improvement, and feedback on the app's usability and ability to support joint activities.

Overall, the study design prioritized allowing participants to autonomously engage with the app without researcher intervention or detailed instructions, enabling the collection of authentic and ecologically valid insights into individual and family-level user experience.

Data Analysis Stage

To analyze the questionnaire responses, we conducted an open–coding–based thematic analysis. Three authors independently reviewed and coded the responses to identify recurring themes and patterns. This collaborative coding process allowed us to reduce individual bias and ensure a robust and reliable interpretation of the data. Discrepancies in coding were discussed and resolved through group discussion until consensus was reached. This approach enabled us to extract meaningful insights about how participants experienced the app and how it influenced their family communication practices.

Findings and Discussion

Our research on Bun Appetit revealed several key insights into how gamified joint activities can facilitate emotional connections between distant family members. Through qualitative analysis of participant feedback, we identified patterns that address our research questions and contribute to the understanding of technology based family communication. Our findings revealed four key themes: (1) Task-Oriented Framing Reduces Initiation Barriers, demonstrating how structured activities lower the barrier of starting remote interactions; (2) Symbolic Companions Anchor Emotional Presence, showing how virtual pets serve as emotional "kinkeepers" that maintain connection; (3) Activity-Based Design Triggers Communication Beyond the Platform, highlighting how shared experiences create natural conversation starters; and (4) Asynchronous Design Supports Sustainable Family Engagement, revealing how flexibility in timing, accommodates busy family schedules. The following sections present our findings organized by thematic areas, examining both the evidence and its relationship to existing literature.

5.1: Task-Oriented Gamified Framework Reduces Initiation Barriers

Our diary study with six families revealed consistent challenges in maintaining meaningful distant communication. Participants specifically described how technology based conversations often felt forced and fizzled out due to lack of shared context. For example, one parent reported, "Sometimes it's hard to get schedules to align, which makes it difficult to plan or follow through with any joint activities. Even when we try to share things like dishes we cook, it often ends up being just a quick photo or message instead of real time interaction." Another noted that before using the app, sharing experiences was often "one sided. I would share something that I did, family would ask questions, I would answer. Sometimes they would respond with "good for you" or "that sounds fun". But it wasn't a shared experience." The structured approach of Bun Appetit, with task based prompts and shared goals, creates clear entry points for interaction that overcome these barriers. One participant observed that "The concept of synchronous joint activities sparked a lot of conversation in text afterwards," indicating how the tasks from the app served as a catalyst for communication beyond the app itself.

Previous research identified a critical challenge in technology based parent-child relationships is discrepancies in expected communication between parents and children [12]. Their systematic review found that "parents and children had different perspectives about the desired frequency, level of reciprocity and disclosure, and mode of communication in spending time together," which led to less meaningful relationships. Similarly, one study found that cooperative gameplay with asymmetric player abilities significantly improved parent-child co-play experiences [5]. In this study of Roblox games, one parent specifically noted, "I really liked the idea of having different roles and the collaboration was really interesting," highlighting how structured activities with clear roles can create more engaging family interactions.

The effectiveness of our task oriented framing extends this research by demonstrating how structured activities can foster meaningful connections between distant family members. One study identified that families often struggle with finding suitable conversation topics, our findings indicate that providing concrete tasks creates natural conversation starters that then go on to extend beyond the platform [12]. Our approach transforms the concept of staying connected into actions that family members can take, which makes emotional connection, a reaction of collaborative activity rather than its goal. This alignment, with established research, reinforces the value of our approach in addressing the challenges of initiating and engaging in meaningful interactions between distant family members.

5.2: Symbolic Companions Anchor Emotional Presence

The virtual pet, Nibbles, emerged as a significant factor in maintaining user engagement and emotional connection. Rather than being a gamification element, Nibbles functioned as an emotional kinkeeper that created a sense of shared presence. One participant explained, “Feeding the bunny was a motivation to not let the tasks go undone for too long. It was satisfying to see the bunny hunger line all green” Another mentioned that checking on Nibbles became a part of their daily routine, serving as a reminder of their family connection even when not actively engaged in cooking tasks.

Research on family communication has established the concept of “kinkeeping” as the emotional and practical work of maintaining family ties. Traditionally, this role often falls to women and creates stress due to the breadth of relationships that must be maintained [4]. However, digital technologies are transforming how kinkeeping occurs with families through virtual spaces. Recent research on family WhatsApp groups in Israel shows how messaging platforms help distribute kinkeeping responsibilities across generations, creating “collective, well-coordinated endeavor” that enhances family belonging [1].

Our findings align with this understanding of digital kinkeeping but introduce a new element: the virtual companion. Nibbles served as both a neutral third party and a shared point of focus that reduced the emotional responsibilities that are associated with kinkeeping and kinkeepers. Caring for Nibbles provided a simplified and low pressure way to maintain connection rather than traditional kinkeeping which can create stress when managing relationships with multiple family members. As one participant expressed, "I like the cute interactions and the ease of use that makes it very impressive to continue using this.” Another participant highlighted how the rabbit served as an emotional anchor: "It was stressful cause I didn't want him to starve", showing how the virtual pet created genuine emotional investment in the shared experience.This suggests that digital companions can serve as effective emotional anchors in joint family applications, particularly when they require shared attention.

5.3: Activity-Based Design Triggers Communication Beyond the Platform

Our findings revealed that although Bun Appetit lacked normal messaging interfaces, participants reported increased communication outside the app. One participant noted that, “the concept of synchronous joint activities sparked a lot of conversation in text afterwards,” demonstrating how the app created natural conversation starters. Another observed that, “getting to see everybody’s photos in one place and react/comment introduced a fun way to spark conversations,” highlighting how the memory sharing feature prompted interactions outside of our app, with family members texting, calling, or video chatting to discuss their cooking progress or compare results without being explicitly prompted to do so by the app.

When comparing to participants’ previous experiences, a prominent contrast emerged in how they communicated before using Bun Appetit. One participant mentioned, “one sided. I would share something that I did, family would ask questions, I would answer. Sometimes they would respond with ‘good for you’ or ‘that sounds fun’. But it wasn’t a shared experience.” Another noted that without structured activities, “Even when we try to share things like dishes we cook, it often ends up being just a quick photo or message instead of a real-time interaction.” Many participants reported that their previous attempts at sharing cooking experiences typically involved sending photos in a family group chat with minimal engagement. One mentioned that, before using the app, joint activity sharing was limited to “photos and videos accompanied with text, “ which they felt was “enough” but not really engaging. Our app introduced task based cooking activities which changed the dynamic of virtual joint activities with a participant noting that this was “my first time doing something like this with [a family member] so it was fun and something new to do.”

Our findings suggest that activity driven designs can transform how families interact despite being separated. Rather than relying on conversation alone, which participants found to be forced, structured activities provided a foundation for more meaningful exchanges within the app and outside of the app as well. These shared experiences became topics of conversation that persisted beyond the app. This can represent a shift from previous communication patterns where participants described interactions as brief. By creating what one participant called “a common experience despite being far away,” Bun Appetit succeeded in transforming ordinary cooking tasks into meaningful shared experiences.

5.4: Asynchronous Design Supports Sustainable Family Engagement

Our study revealed that asynchronous game design successfully accommodates the varied schedules and time constraints of family members, enabling sustained engagement over time. Participants appreciated the flexibility to engage with the application on their own terms. One parent mentioned, “I communicated with them much more often than I otherwise would.” Another participant noted, “The concept of synchronous joint activities sparked a lot of conversation in text afterwards” The asynchronous nature of the application allowed families to maintain connection despite not being in each other’s presences.

Research on family interactions suggests that asynchronous communication can help overcome barriers to shared activities, particularly when family members have varying schedules. “For families, where people can have vastly different availability, motivations and skills, opportunities for playing together can be scarce” [10]. Their work revealed that asynchronous design where players don’t need to play at the same time nor place, can create meaningful interdependent experiences while still accommodating to the varying schedules between family members. This approach enables both adults and children to interact meaningfully within the constraints of their unique daily lives.

Our findings align with these insights, showing that asynchronous interaction facilitated sustained engagement by removing the time related barriers that typically prevent families from participating in distant, joint activities. Several participants commented on how they were able to integrate their application into their existing routines, such as checking on Nibbles or completing tasks during brief moments throughout the day. The presence of a virtual pet created a persistent checkpoint that family members could engage with independently while still feeling connected to each other. The design approach enabled families to maintain connection without requiring the in-person presence of each other, creating what one participant in Pais et al.’s study described as a “communal” experience that adapts to each member’s availability rather than forcing synchronization, which would result in fake interaction. By allowing independent interaction with independent outcomes, Bun Appetit supported sustainable engagement patterns that could fit into the complex reality of modern, distant families.

Conclusion

In an era where distance often separates families, maintaining meaningful connections is both increasingly important and challenging. This study explored how everyday activities, such as cooking, can be transformed into collaborative, emotionally engaging experiences through a gamified mobile application. By designing and evaluating *Bun Appétit*, we investigated how task-oriented design, asynchronous interaction, and the presence of a virtual companion can help reduce communication barriers and foster emotional connection among family members living apart. Our user study provided preliminary evidence that gamified experiences can extend beyond simple exchanges of messages or photos to create moments of emotional closeness, particularly between parents and children. Participants reported that the playful framing of routine tasks increased motivation, sustained engagement, and created opportunities for meaningful family interaction. These findings offer promising directions for the design of future family-centered technologies and contribute to ongoing conversations in the CHI PLAY and HCI communities about how to support relational maintenance and emotional connection through play.

We acknowledge several limitations of this study. First, the sample size was relatively small, and participants were recruited from the authors’ networks, which may limit the generalizability of the findings. Second, the study period was short (four days), providing only a snapshot of engagement, rather than insights into long-term use and retention. For future work, we plan to conduct longitudinal studies with a more diverse pool of families to examine how playful, collaborative activities influence communication patterns over time and across different cultural contexts. Additionally, future research may explore the integration of adaptive or personalized elements to further tailor the experience to family members' individual needs and preferences.

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